Judson, *Abstract Algebra: Theory and Applications* Chapter 10 (Normal Subgroups and Factor Groups): Solutions to Exercises 1–15

Quick checks used throughout

- A subgroup of index 2 is always normal (Exercise 10).
- Every subgroup of an abelian group is normal.
- A simple group has no normal subgroups besides $\{e\}$ and itself.
- A Sylow p-subgroup is normal if and only if it is the only Sylow p-subgroup.
- Otherwise, find $g \in G$ and $h \in H$ with $ghg^{-1} \notin H$. For permutations, $g(a\ b\ c)g^{-1} = (g(a)\ g(b)\ g(c))$.

Exercise 1

For each of the following groups G, determine whether H is a normal subgroup of G. If H is a normal subgroup, write out a Cayley table for the factor group G/H.

(a) $G = S_4$ and $H = A_4$

(b) $G = A_5$ and $H = \{(1), (123), (132)\}$

(c) $G = S_4$ and $H = D_4$

(d) $G = Q_8$ and $H = \{1, -1, I, -I\}$

(e) $G = \mathbb{Z}$ and $H = 5\mathbb{Z}$

Solution

(a) H is normal, since $[S_4 : A_4] = 2$. (Equivalently, $A_4 = \ker(\text{sgn})$.) Thus $G/H \cong \mathbb{Z}_2$, with cosets H (even permutations) and $(12)H$ (odd permutations).

$\cdot$	H	$(12)H$
H	H	$(12)H$
$(12)H$	$(12)H$	H

(b) H is not normal. Let $g = (124) \in A_5$. Then

$$g(123)g^{-1} = \big(g(1)\ g(2)\ g(3)\big) = (243) \notin H.$$

(Alternatively, A_5 is simple, so its only normal subgroups are $\{(1)\}$ and A_5; H is neither.)

(c) Take $D_4 = \langle (1234), (13) \rangle$, which has the eight elements

$$(1),\ (1234),\ (13)(24),\ (1432),\ (13),\ (24),\ (12)(34),\ (14)(23).$$

Let $g = (12)$. Then $g(13)g^{-1} = (g(1)\ g(3)) = (23) \notin D_4$, so H is not normal. (Since $|D_4| = 8$ is the largest power of 2 dividing $|S_4| = 24$, D_4 is a Sylow 2-subgroup, and there are three of them, so none is normal.)

(d) H is normal, since $[Q_8 : H] = 2$. Thus $G/H \cong \mathbb{Z}_2$, with cosets H and $JH = \{J, -J, K, -K\}$. Note $J^2 = -1 \in H$, so $(JH)(JH) = H$.

$\cdot$	H	JH
H	H	JH
JH	JH	H

(e) H is normal, since $\mathbb{Z}$ is abelian. Thus $G/H = \mathbb{Z}/5\mathbb{Z} \cong \mathbb{Z}_5$. Writing $[a] = a + 5\mathbb{Z}$, the operation is $[a] + [b] = [a + b]$:

$+$	[0]	[1]	[2]	[3]	[4]
[0]	[0]	[1]	[2]	[3]	[4]
[1]	[1]	[2]	[3]	[4]	[0]
[2]	[2]	[3]	[4]	[0]	[1]
[3]	[3]	[4]	[0]	[1]	[2]
[4]	[4]	[0]	[1]	[2]	[3]

Summary. H is normal in (a), (d), (e), and not normal in (b), (c).

Exercise 2

Find all the subgroups of D_4. Which subgroups are normal? What are all the factor groups of D_4 up to isomorphism?

Solution

Write $D_4 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1} \rangle$, so

$$D_4 = \{1, r, r^2, r^3, s, sr, sr^2, sr^3\}, \qquad rs = sr^{-1}.$$

The elements of order 2 are r^2, s, sr, sr^2, sr^3; the elements of order 4 are r, r^3.

By Lagrange, subgroups have order $1, 2, 4$, or 8.

- **Order 1 and 8:** $\{1\}$ and D_4.
- **Order 2:** one for each element of order 2: $\langle r^2 \rangle,\ \langle s \rangle,\ \langle sr \rangle,\ \langle sr^2 \rangle,\ \langle sr^3 \rangle$.
- **Order 4:** the cyclic subgroup $\langle r \rangle = \{1, r, r^2, r^3\}$ (the only cyclic one, since r, r^3 are the only elements of order 4), and two Klein four-groups

  $$V_1 = \{1, r^2, s, sr^2\}, \qquad V_2 = \{1, r^2, sr, sr^3\}.$$

  (In a Klein four-group the product of two reflections sr^i, sr^j is a rotation r^{j-i} of order 2, so $j - i \equiv 2 \pmod 4$.)

So D_4 has 10 subgroups in total.

Normality.

- $\{1\}$ and D_4 are normal.
- $\langle r^2\rangle = Z(D_4)$ is normal.
- $\langle r\rangle$, V_1, V_2 have index 2, so they are normal.
- $\langle s\rangle$ is not normal: $rsr^{-1} = sr^{-2} = sr^2 \notin \langle s\rangle$. Similarly $\langle sr^2\rangle$ is not normal. Also $r(sr)r^{-1} = rs = sr^3 \notin \langle sr\rangle$, so $\langle sr\rangle$ is not normal, and likewise $\langle sr^3\rangle$.

Factor groups.

- $D_4/\{1\} \cong D_4$.
- $D_4/\langle r^2\rangle$ has order 4. Every element of D_4 squares into $\{1, r^2\}$, so every nonidentity coset squares to the identity coset. Hence $D_4/\langle r^2\rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.
- $D_4/\langle r\rangle$, D_4/V_1, D_4/V_2 each have order 2, so each is isomorphic to $\mathbb{Z}_2$.
- D_4/D_4 is the trivial group.

Up to isomorphism, the factor groups are $D_4,\ \mathbb{Z}_2 \times \mathbb{Z}_2,\ \mathbb{Z}_2$, and the trivial group.

Exercise 3

Find all the subgroups of the quaternion group, Q_8. Which subgroups are normal? What are all the factor groups of Q_8 up to isomorphism?

Solution

$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$. The element 1 has order 1, -1 has order 2, and $\pm i, \pm j, \pm k$ all have order 4.

A subgroup containing an element of order 4, say i, contains $\langle i\rangle = \{1, i, -1, -i\}$, which has order 4; by Lagrange it is $\langle i\rangle$ or Q_8. The same holds for j and k. A subgroup with no element of order 4 lies in $\{1, -1\}$. Hence the subgroups are

$$\{1\}, \quad \{\pm 1\}, \quad \langle i\rangle, \quad \langle j\rangle, \quad \langle k\rangle, \quad Q_8.$$

Normality. All six are normal: $\{1\}$ and Q_8 trivially, $\{\pm 1\} = Z(Q_8)$, and $\langle i\rangle, \langle j\rangle, \langle k\rangle$ have index 2.

Factor groups.

- $Q_8/\{1\} \cong Q_8$.
- $Q_8/\{\pm 1\}$ has order 4. Every $x \in Q_8$ satisfies $x^2 \in \{\pm 1\}$, so every coset squares to the identity coset. Hence $Q_8/\{\pm 1\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.
- $Q_8/\langle i\rangle \cong Q_8/\langle j\rangle \cong Q_8/\langle k\rangle \cong \mathbb{Z}_2$.
- Q_8/Q_8 is trivial.

Up to isomorphism, the factor groups are $Q_8,\ \mathbb{Z}_2 \times \mathbb{Z}_2,\ \mathbb{Z}_2$, and the trivial group.

Exercise 4

Let T be the group of nonsingular upper triangular 2×2 matrices with entries in $\mathbb{R}$; that is, matrices of the form

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix},$$

where $a, b, c \in \mathbb{R}$ and $ac \neq 0$. Let U consist of matrices of the form

$$\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix},$$

where $x \in \mathbb{R}$.

(a) Show that U is a subgroup of T.

(b) Prove that U is abelian.

(c) Prove that U is normal in T.

(d) Show that T/U is abelian.

(e) Is T normal in $\mathrm{GL}_2(\mathbb{R})$?

Solution

Write $u(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$. Note that

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}\begin{pmatrix} a' & b' \\ 0 & c' \end{pmatrix} = \begin{pmatrix} aa' & ab'+bc' \\ 0 & cc' \end{pmatrix}. \tag{$*$}$$

(a) $U \subseteq T$, and $u(0) = I \in U$. By $(*)$, $u(x)u(y) = u(x+y) \in U$. Also $u(x)u(-x) = u(0) = I$, so $u(x)^{-1} = u(-x) \in U$. Hence U is a subgroup of T.

(b) $u(x)u(y) = u(x+y) = u(y+x) = u(y)u(x)$, so U is abelian.

(c) Let $g = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in T$, so $g^{-1} = \begin{pmatrix} 1/a & -b/(ac) \\ 0 & 1/c \end{pmatrix}$. Then

$$g\,u(x) = \begin{pmatrix} a & ax+b \\ 0 & c \end{pmatrix}, \qquad g\,u(x)\,g^{-1} = \begin{pmatrix} 1 & -\dfrac{b}{c} + \dfrac{ax+b}{c} \\ 0 & 1 \end{pmatrix} = u\Big(\frac{ax}{c}\Big) \in U.$$

So $gUg^{-1} \subseteq U$ for all $g \in T$, and U is normal in T.

(d) Let $g = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$ and $h = \begin{pmatrix} a' & b' \\ 0 & c' \end{pmatrix}$ be in T. By $(*)$, gh and hg both have diagonal entries aa' and cc'. Therefore $(hg)^{-1}(gh)$ is upper triangular with diagonal entries $1, 1$, so it lies in U. Hence $ghU = hgU$, that is,

$$(gU)(hU) = (hU)(gU),$$

and T/U is abelian.

(e) No. Take $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in T$ and $g = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{R})$, where $g^{-1} = g$. Then

$$gAg^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \notin T.$$

So T is not normal in $\mathrm{GL}_2(\mathbb{R})$.

Exercise 5

Show that the intersection of two normal subgroups is a normal subgroup.

Solution

Let N and M be normal subgroups of G. The intersection of two subgroups is a subgroup, so $N \cap M \leq G$. Let $g \in G$ and $x \in N \cap M$. Since $x \in N$ and N is normal, $gxg^{-1} \in N$; since $x \in M$ and M is normal, $gxg^{-1} \in M$. Hence $gxg^{-1} \in N \cap M$, so $N \cap M$ is normal in G. $\square$

Exercise 6

If G is abelian, prove that G/H must also be abelian.

Solution

Since G is abelian, H is normal, so G/H is a group. For $a, b \in G$,

$$(aH)(bH) = (ab)H = (ba)H = (bH)(aH). \quad \square$$

Exercise 7

Prove or disprove: If H is a normal subgroup of G such that H and G/H are abelian, then G is abelian.

Solution

This is false. Let $G = S_3$ and $H = A_3 = \{(1), (123), (132)\}$. Then $H \cong \mathbb{Z}_3$ is abelian and normal (index 2), and $G/H \cong \mathbb{Z}_2$ is abelian. But S_3 is not abelian, since $(12)(13) \neq (13)(12)$.

Exercise 8

If G is cyclic, prove that G/H must also be cyclic.

Solution

Let $G = \langle g \rangle$. Then G is abelian, so H is normal and G/H is a group. Any element of G/H has the form $g^n H = (gH)^n$ for some n. Hence $G/H = \langle gH \rangle$ is cyclic. $\square$

Exercise 9

Prove or disprove: If H and G/H are cyclic, then G is cyclic.

Solution

This is false. Let $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ and $H = \langle (1, 0) \rangle \cong \mathbb{Z}_2$. Then H is cyclic, and G/H has order 2, so it is cyclic. But G is not cyclic: it has order 4 and every element has order at most 2, so no element generates G.

Exercise 10

Let H be a subgroup of index 2 of a group G. Prove that H must be a normal subgroup of G. Conclude that S_n is not simple.

Solution

Since $[G : H] = 2$, there are exactly two left cosets, and they partition G. One is H, so the other is $G \setminus H$. The same holds for right cosets.

Let $g \in G$.

- If $g \in H$, then $gH = H = Hg$.
- If $g \notin H$, then $gH \neq H$, so $gH = G \setminus H$. Likewise $Hg = G \setminus H$. So $gH = Hg$.

Thus $gH = Hg$ for all $g \in G$, so H is normal. □

Conclusion. For $n \geq 3$, the alternating group A_n has index 2 in S_n, so it is normal by the above. It is a proper subgroup, and it is nontrivial since $(123) \in A_n$. Hence S_n has a normal subgroup other than $\{(1)\}$ and S_n, so S_n is not simple for $n \geq 3$. (For $n = 2$, $S_2 \cong \mathbb{Z}_2$ is simple.)

Exercise 11

Let G be a group of order p^2, where p is a prime number. If H is a subgroup of G of order p, show that H is normal in G. Prove that G must be abelian.

Solution

H is normal. Let G act on the set G/H of left cosets (there are p of them) by left multiplication, $g \cdot xH = gxH$. This gives a homomorphism $\varphi : G \to S_p$. Let $K = \ker \varphi$.

If $g \in K$, then $gH = H$, so $g \in H$. Thus $K \subseteq H$ and $|K| \in \{1, p\}$.

By the First Isomorphism Theorem, $G/K \cong \varphi(G) \leq S_p$, so $|G/K| = p^2/|K|$ divides $p!$. The prime p divides $p!$ exactly once, so $p^2 \nmid p!$. Since $|G/K|$ is a power of p, we get $|G/K| \leq p$, hence $|K| \geq p$. Therefore $|K| = p$ and $K = H$. A kernel is normal, so $H \trianglelefteq G$.

G is abelian. Elements of G have order 1, p, or p^2. If some element has order p^2, then G is cyclic, hence abelian. Otherwise every non-identity element has order p.

Let $a, b \in G$ with $a \neq e$ and $b \notin \langle a \rangle$ (if $b \in \langle a \rangle$ then a and b certainly commute). Put $H = \langle a \rangle$ and $K = \langle b \rangle$. Both have order p, so both are normal by the first part. They are distinct subgroups of prime order, so $H \cap K = \{e\}$. Now

$$aba^{-1}b^{-1} = (aba^{-1})b^{-1} \in K \quad \text{(since } K \text{ is normal)},$$

$$aba^{-1}b^{-1} = a(ba^{-1}b^{-1}) \in H \quad \text{(since } H \text{ is normal)}.$$

So $aba^{-1}b^{-1} \in H \cap K = \{e\}$, which gives $ab = ba$. Hence any two elements of G commute, and G is abelian. □

Exercise 12

If a group G has exactly one subgroup H of order k, prove that H is normal in G.

Solution

Let $g \in G$. The set gHg^{-1} is a subgroup of G: it contains $e = geg^{-1}$, and $(ghg^{-1})(gh'g^{-1}) = g(hh')g^{-1}$ and $(ghg^{-1})^{-1} = gh^{-1}g^{-1}$ both lie in it. The map $h \mapsto ghg^{-1}$ is a bijection $H \to gHg^{-1}$, so $|gHg^{-1}| = k$. Since H is the only subgroup of order k, $gHg^{-1} = H$. As g was arbitrary, H is normal. □

Exercise 13

Define the centralizer of an element g in a group G to be the set

$$C(g) = \{x \in G : xg = gx\}.$$

Show that $C(g)$ is a subgroup of G. If g generates a normal subgroup of G, prove that $C(g)$ is normal in G.

Solution

Subgroup. Clearly $e \in C(g)$. If $x, y \in C(g)$, then $(xy)g = x(yg) = x(gy) = (xg)y = (gx)y = g(xy)$, so $xy \in C(g)$. If $x \in C(g)$, multiply $xg = gx$ on the left and right by x^{-1} to get $gx^{-1} = x^{-1}g$, so $x^{-1} \in C(g)$.

Normality. Suppose $\langle g \rangle \trianglelefteq G$. Let $h \in G$ and $x \in C(g)$. Since $\langle g \rangle$ is normal, $h^{-1}gh = g^m$ for some integer m. Since x commutes with g, it also commutes with g^m. Then

$$\begin{aligned}(hxh^{-1})\, g\, (hxh^{-1})^{-1} &= hx\,(h^{-1}gh)\,x^{-1}h^{-1} \\ &= hx\, g^m\, x^{-1}h^{-1} \\ &= h\, g^m\, h^{-1} \\ &= h\,(h^{-1}gh)\,h^{-1} = g.\end{aligned}$$

Hence $(hxh^{-1})g = g(hxh^{-1})$, so $hxh^{-1} \in C(g)$. Therefore $C(g)$ is normal in G. □

Exercise 14

Recall that the center of a group G is the set

$$Z(G) = \{x \in G : xg = gx \text{ for all } g \in G\}.$$

(a) Calculate the center of S_3.

(b) Calculate the center of $\mathrm{GL}_2(\mathbb{R})$.

(c) Show that the center of any group G is a normal subgroup of G.

(d) If $G/Z(G)$ is cyclic, show that G is abelian.

Solution

(a) $Z(S_3) = \{(1)\}$. The identity is central. Conjugation by (12) sends (123) to $(g(1)\,g(2)\,g(3)) = (213) = (132) \neq (123)$, so neither (123) nor (132) is central. Conjugation by (123) sends (12) to (23), (13) to (12), and (23) to (13), so no transposition is central.

(b) $Z(\mathrm{GL}_2(\mathbb{R})) = \{\lambda I : \lambda \in \mathbb{R},\ \lambda \neq 0\}$.

Scalar matrices commute with every matrix, so they are central. Conversely, let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be central. Then A commutes with $B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and with $C = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. Now

$$AB = \begin{pmatrix} a & a+b \\ c & c+d \end{pmatrix}, \qquad BA = \begin{pmatrix} a+c & b+d \\ c & d \end{pmatrix},$$

and $AB = BA$ gives $c = 0$ and $a = d$. Also

$$AC = \begin{pmatrix} a+b & b \\ c+d & d \end{pmatrix}, \qquad CA = \begin{pmatrix} a & b \\ a+c & b+d \end{pmatrix},$$

and $AC = CA$ gives $b = 0$. So $A = aI$, with $a \neq 0$ because A is invertible.

(c) $Z(G)$ is a subgroup: $e \in Z(G)$; if $x, y \in Z(G)$ and $g \in G$, then $(xy)g = x(yg) = x(gy) = (xg)y = g(xy)$; and from $xg = gx$ we get $gx^{-1} = x^{-1}g$ as in Exercise 13. It is normal because for $z \in Z(G)$ and $g \in G$, $gzg^{-1} = zgg^{-1} = z \in Z(G)$.

(d) Suppose $G/Z(G)$ is cyclic, say $G/Z(G) = \langle gZ(G)\rangle$. Let $x, y \in G$. Then $xZ(G) = g^iZ(G)$ and $yZ(G) = g^jZ(G)$ for some i, j, so $x = g^iz$ and $y = g^jw$ with $z, w \in Z(G)$. Since z, w are central,

$$xy = g^izg^jw = g^{i+j}zw, \qquad yx = g^jwg^iz = g^{i+j}wz = g^{i+j}zw.$$

So $xy = yx$, and G is abelian. □

Exercise 15

Let G be a group and let $G' = \langle aba^{-1}b^{-1}\rangle$; that is, G' is the subgroup of all finite products of elements in G of the form $aba^{-1}b^{-1}$. The subgroup G' is called the *commutator subgroup* of G.

(a) Show that G' is a normal subgroup of G.

(b) Let N be a normal subgroup of G. Prove that G/N is abelian if and only if N contains the commutator subgroup of G.

Solution

(a) For $g, a, b \in G$,

$$g(aba^{-1}b^{-1})g^{-1} = (gag^{-1})(gbg^{-1})(gag^{-1})^{-1}(gbg^{-1})^{-1},$$

which is again a commutator. (Note also that the inverse of a commutator, $(aba^{-1}b^{-1})^{-1} = bab^{-1}a^{-1}$, is a commutator, so G' consists exactly of the finite products of commutators.) If $x = c_1c_2\cdots c_n$ is a product of commutators, then

$$gxg^{-1} = (gc_1g^{-1})(gc_2g^{-1})\cdots(gc_ng^{-1})$$

is again a product of commutators. So $gG'g^{-1} \subseteq G'$ for all $g \in G$, and G' is normal.

(b) ($\Rightarrow$) Suppose G/N is abelian. For any $a, b \in G$,

$$(aba^{-1}b^{-1})N = (aN)(bN)(aN)^{-1}(bN)^{-1} = N,$$

since G/N is abelian. So $aba^{-1}b^{-1} \in N$. Thus N contains every commutator, and since N is a subgroup it contains all finite products of them, so $G' \subseteq N$.

($\Leftarrow$) Suppose $G' \subseteq N$. For any $a, b \in G$, $aba^{-1}b^{-1} \in G' \subseteq N$, so

$$(aN)(bN)(aN)^{-1}(bN)^{-1} = (aba^{-1}b^{-1})N = N,$$

the identity of G/N. Hence $(aN)(bN) = (bN)(aN)$, and G/N is abelian. $\square$